

Roger Cooke's contribution to the Discussion of Bayesian Statistics in Scientific Induction A Decision Analysis Perspective by Simon French

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These lightning comments are addressed to the Model Scientist (MS). They draw on my experience as Lead Author for Risk and Uncertainty of the IPCC's 5th Assessment Report (AR) (Kunreuther et al 2014), reported in (Cooke 2015). First, I wish to acknowledge Simon's invaluable early work in deflecting "alternative uncertainties" which contributed greatly to the following table.

Table 1 Word fragment counts presented at the conference Uncertainty in Artificial Intelligence, 1985, 2000 and 2012 (Nature Climate Change 5, 8 -10, 2015)

	1985	2000	2012
Fuzzy	20%	9%	1%
Belief function	29%	1%	0%
Possibilistic	0%	11%	0%
Certainty factor	20%	0%	0%
Imprecise prob	0%	1%	1%
Random sets	0%	0%	1%
Non-monotonic	5%	0%	0%
Bayes	26%	78%	97%

"Fast thinkers" aren't the ones requiring MS's gentle ministrations. IPCC 4th AR scientists and communicators at the top of the foodchain reason as if high confidence in each of several statements entails high confidence in all, fail to distinguish conditional and joint probability and think that 99% chance is "virtually certain". The IPCC's 5th AR states: "Natural language is not adequate for propagating and communicating uncertainty" (Box 2.2). Retaining quantitative likelihood statements, it swaps the quantitative confidence levels for "qualitative syntheses", which brings us to MS's embrace of Knight's unquantifiable uncertainties. Knight wrote: "We can also employ the terms 'objective' and 'subjective' probability to designate the risk and uncertainty." (Knight 1921, III.VIII.1). Knight in 1921 didn't know how to quantify partial belief. We do (Ramsey 1931) and MS can remove the Knightian fig leaf over unquantified uncertainties. Knight nonetheless anticipated the objective validation of expert subjective probabilities (Knight, 1921, III VIII.43).

David Archer (2009, p.95) writes "A climate change of … 5~6°C, would be catastrophic to human civilization." According to the IPCC AR5, Equilibrium Climate Sensitivity is "…extremely unlikely [< 5% chance] less than 1°C (high confidence) and very unlikely [< 10% chance] greater than 6°C (medium confidence). " (Collins et al 2013, p.1111). "Medium confidence" previously meant "about a 5 out of 10 chance". Now it means…? At least it means that the statement "Prob(ECS >6°C) < 10%" has declarative meaning, they can't have medium confidence in a meaningless string of words. People must assign it some degree of belief. How much? Would the IPCC authors have high confidence that the probability that ECS exceeds 11°C is less than 10%? Whom can we ask? More to the point, would MS put our grandkids on an airplane in which authorities had medium confidence that the probability of a crash was less than 10%? I implore MS to evaluate IPCC's communication of uncertainty, its not unimportant.

Kunreuther H., S. Gupta, V. Bosetti, R. Cooke, V. Dutt, M. Ha-Duong, H. Held, J. Llanes-Regueiro, A. Patt, E. Shittu, and E.Weber, (2014): Integrated Risk and Uncertainty Assessment of Climate Change Response Policies. In: Climate Change 2014:Mitigation of Climate Change. Contribution of Working Group III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change [Edenhofer, O., R. Pichs-Madruga, Y. Sokona, E. Farahani, S. Kadner, K. Seyboth, A. Adler, I. Baum, S. Brunner, P. Eickemeier, B. Kriemann, J. Savolainen, S. Schlömer, C. von Stechow, T. Zwickel and J.C. Minx (eds.)]. Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA.

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